



PORTFOLIO PROJECT

GeneSight

AI Research Assistant for Genomics

Ask a research question. Get ranked, validated hypotheses backed by 33 million+ scientific papers — in minutes.

FastAPI

OpenAI GPT-4o

Neo4j Graph DB

LangGraph

Clerk Auth

Vercel

THE PROBLEM

Research is drowning in data.

Information overload

33M+ papers on PubMed — 4,000 new ones added every single day.

Weeks, not hours

Manual literature reviews take months before a hypothesis even forms.

Missed connections

Critical gene-disease links buried across thousands of disconnected papers.

33M+

Scientific Papers

on PubMed alone

GeneSight reads what humans can't.

Three steps to a breakthrough.

From question to insight in under 5 minutes.

01 You Ask

"BRCA1 and breast cancer risk?"

Type any genomics question in plain language. No special syntax needed.

02 AI Analyzes

8 agents search → extract → cross-check

Literature search, entity extraction, gene enrichment, and knowledge graph — all in parallel.

03 You Discover

5 ranked hypotheses + evidence

Get confidence-scored hypotheses, supporting papers, and a visual knowledge graph.

The 8-Agent AI Pipeline

Every query runs through 8 specialised AI agents working in sequence and in parallel.



What researchers actually see.

Hypothesis #1

87% Confidence

✓ Strong

"BRCA1 deficiency activates alternative end-joining pathways that increase genomic instability in breast cancer cells."

Genes

Papers

Pathway

BRCA1, TP53, RAD51

23 supporting

DNA Damage Repair

Evidence strength

Also included:



Knowledge Graph

Interactive D3 gene-disease-protein network



Cited Papers

Every paper linked to the hypothesis that supports it



Quality Score

0-100 overall analysis score with per-agent trace

Built with the best tools for the job.

AI & Intelligence

OpenAI GPT-4o

Hypothesis generation & validation

LangChain / LangGraph

Agent orchestration framework

Biopython Entrez

PubMed literature retrieval API

Data & Storage

Neo4j AuraDB

Knowledge graph — genes, diseases, proteins

Neon PostgreSQL

User history, rate limits, auth state

NCBI Gene + UniProt

Biological databases enrichment

API & Infrastructure

FastAPI + Uvicorn

Async Python REST API

Clerk Auth (RS256)

JWT auth, JWKS-cached, SSR-safe

Vercel

Frontend CDN & serverless deployment

Production-grade, not just a demo.



Prompt Injection Guard

Blocks adversarial inputs before they reach any LLM call.



Hallucination Check

Every gene verified against NCBI Gene database in real-time.



Evidence Validation

GPT-4o reads each paper — rejects unsupported hypotheses.



Circuit Breakers

Auto-recover from OpenAI / NCBI / Neo4j failures. No cascades.



Auth + Rate Limiting

Clerk RS256 JWT with JWKS caching. 10 req/min IP throttle.



Smart Caching

Identical queries return in <10ms. No duplicate LLM spend.

BY THE NUMBERS

Scale that speaks for itself.

33M+

Papers searchable

via NCBI Entrez API

8

AI agents

in the async pipeline

< 5

Minutes

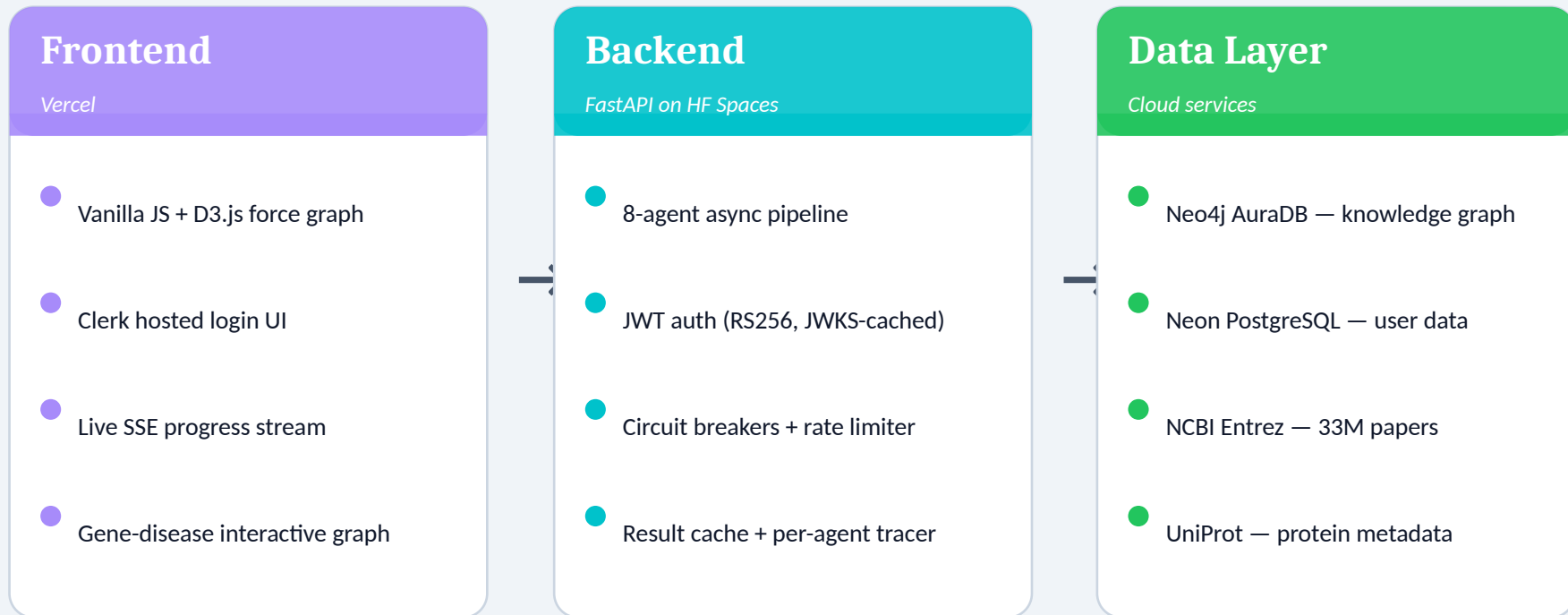
end-to-end per query

30

Parallel requests

NCBI + UniProt, same gather

Full-stack architecture overview.



Built by Akshita.

GeneSight is a full-stack AI system — async FastAPI, 8-agent LLM pipeline, Neo4j knowledge graph, and a live Vercel frontend. Production-grade from day one.

Live App

gene-sight-seven.vercel.app

GitHub

github.com/akshita270/GeneSight

Contact

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"Turn 33 million papers into one clear answer."